UNIT 3 · CIRCULAR MOTION & GRAVITATION · WEEK 1

Problem Set 10 — Circular Motion & Gravitation

Physics 2026–27 · due at the start of class, Monday, January 11 · Unit 3 Test Thu, Jan 14

NAME _____ DATE _____

How to work this set. Equations this week: $a_c = v^2/r$ (always toward the center), $v = 2\pi r/T$, $F_c = mv^2/r$, and $F_g = Gm_1m_2/r^2$ with $G = 6.674 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2$. Remember: "centripetal force" is a *label* for whatever real force points centerward — it never gets its own arrow on an FBD. Scientific notation is back; use ≈ honestly.

1 Constant speed, still accelerating

CONCEPT

a. A car circles a roundabout at a perfectly steady 20 km/h. Its speedometer never moves — yet it is accelerating. Explain how both can be true.

b. True or false, with one sentence: a passenger in that car is pushed outward by a "centrifugal force."

ANSWERS

2 Straight to the formula

A_C

A cyclist rounds a curve at `9.0 m/s` with a turning radius of `4.5 m`. Run KDEA for the centripetal acceleration — magnitude and direction.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	$A_C = V^2/R$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

3 Ball on a string

PERIOD → FORCE

A `0.20 kg` ball swings in a horizontal circle of radius `0.75 m`, completing one lap every `0.60 s`.

a. Find its speed. (`v = 2πr/T`)

b. Find its centripetal acceleration.

c. Find the net centripetal force — and name the real force that supplies it.

K	EVERY GIVEN, WITH UNITS
D	THREE TARGETS, THREE UNITS
E	$V = 2\Pi R/T \cdot A_C = V^2/R \cdot F_C = MA_C$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY · ≈ WHERE Π FORCES IT

4 The flat curve

FRICTION AS F_C

An `1100 kg` car takes a flat curve of radius `45 m` at `14 m/s`.

a. How much centripetal force does the turn require?

b. What minimum coefficient of static friction keeps the car on the road? (Set $\mu_s mg = mv^2/r$ and watch the mass cancel.)

c. Why μ_s and not μ_k? One sentence — a Workshop I subscript question, three months later.

K	EVERY GIVEN, WITH UNITS
D	THE TARGETS & THEIR UNITS (M HAS NONE — SAY SO)
E	$F_C = MV^2/R \cdot M_s = V^2/(RG)$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 Inverse-square reasoning

GRAVITATION

No calculator for this one — pure proportional reasoning from $F_g = Gm_1m_2/r^2$:

a. The distance between two masses doubles. What happens to F_g?

b. The distance shrinks to one-third of the original. What happens to F_g?

c. Both masses double *and* the distance doubles. Now what?

REASONING

6 Earth pulls the satellite

$F_G = GM_1M_2/R^2$

A $600\ kg$ satellite orbits at $r = 7.0 \times 10^6\ m$ from Earth's center. ($M_{earth} = 5.97 \times 10^{24}\ kg$.) Run KDEA for the gravitational force on it — this is the scientific-notation workout Unit 0 trained you for, so track the exponents in your A row.

K	EVERY GIVEN, WITH UNITS, IN SCIENTIFIC NOTATION
D	THE TARGET & ITS UNIT
E	$F_G = GMM/R^2$
A	SYMBOLS FIRST · EXPONENT BOOKKEEPING SHOWN

7 Name the centripetal force

SOURCES

For each, name the *real* force (or forces) playing the centripetal role — "centripetal force" is not an acceptable answer:

a. The Moon orbiting Earth.

b. A coin resting on a spinning turntable.

c. A roller-coaster car at the very top of a vertical loop. (Two forces — both point the same way there.)

d. A car on a banked, frictionless curve.

ANSWERS

*Graded on the homework 4–3–2–1 scale. Last problem set of Semester 1 — the **Unit 3 Test (Thu, Jan 14)** covers this material plus orbits.*